

Partial Fraction Decomposition

Find the partial fraction decomposition of each.

1) $\frac{-5x+4}{x^2-x}$

$$-\frac{4}{x} - \frac{1}{x-1}$$

2) $\frac{3x+10}{x^2+9x+20}$

$$-\frac{2}{x+4} + \frac{5}{x+5}$$

3) $\frac{-2x^2+4x+14}{x^2-6x+5}$

$$-2 - \frac{4}{x-5} - \frac{4}{x-1}$$

4) $\frac{2x^2-9x-10}{x^2-5x}$

$$2 + \frac{2}{x} - \frac{1}{x-5}$$

5) $\frac{-7x-15}{x^2+6x+9}$

$$-\frac{7}{x+3} + \frac{6}{(x+3)^2}$$

6) $\frac{-2x^2+19x-13}{x^3-7x^2+11x-5}$

$$\frac{2}{x-5} - \frac{4}{x-1} - \frac{1}{(x-1)^2}$$

7) $\frac{-6x^2+3x+5}{x^3-x}$

$$-\frac{5}{x} - \frac{2}{x+1} + \frac{1}{x-1}$$

8) $\frac{20x+9}{25x^2+20x+4}$

$$\frac{4}{5x+2} + \frac{1}{(5x+2)^2}$$

9) $\frac{-4x^4-26x^2-2x^3-8x-44}{(x+1)(x^2+3)^2}$

$$-\frac{4}{x+1} - \frac{2}{x^2+3} - \frac{2}{(x^2+3)^2}$$

10) $\frac{-2x^3+36x^2-199x+375}{x(x-5)^3}$

$$-\frac{3}{x} + \frac{1}{x-5} + \frac{1}{(x-5)^2} + \frac{6}{(x-5)^3}$$

11) $\frac{15x^2-11x-5}{x(x+1)(2x-5)}$

$$\frac{1}{x} + \frac{3}{x+1} + \frac{7}{2x-5}$$

12) $\frac{2x^4-8x^2-10+3x^3-9x}{x(x^2+1)(x^2-5)}$

$$\frac{2}{x} + \frac{2}{x^2+1} + \frac{1}{x^2-5}$$